

Gaussian Interpretation of Quantum mechanics

A Gauss-Style Lens on Quantum Notation

The goal of this note is modest: to offer a conceptual lens that some readers may find clarifying when thinking about the symbols of quantum mechanics. The perspective is not presented as an ontology or a competing interpretation, but simply as a structurally coherent way of looking at familiar notation. It draws inspiration from a classical idea that long predates quantum theory: Gauss's observation that the interior of a region is never accessed directly, while the boundary is the only place where information becomes available.

Throughout this note, the term **boundary** refers to the degrees of freedom accessible to an observer—whatever aspects of a system can, in principle, be measured or interacted with. The **interior** refers to the degrees of freedom that remain inaccessible. No spatial surface is implied; the distinction is informational rather than geometric.

In Gauss's law, the observer does not inspect the charge distribution inside a surface. Instead, the outward flux through the boundary is measured, and the interior is inferred from that boundary data. This asymmetry—interior hidden, boundary accessible—turns out to be a surprisingly helpful template for understanding the operational roles of kets, bras, amplitudes, and probabilities in quantum mechanics. Nothing in this analogy changes the formalism; it simply highlights a structural pattern that is already present.

Readers are invited to treat this as a way of seeing rather than a claim about what ultimately exists. The analogy is optional, but it can make several features of the quantum formalism feel less mysterious: superposition becomes boundary ambiguity, collapse becomes boundary commitment, and entanglement becomes a shared boundary constraint. The sections below develop this viewpoint step by step, showing how a Gauss-style reading of quantum notation can serve as a compact and intuitive guide to the mathematics.

Gauss's law is expressed by the integral identity

$$\iint_{\partial R} \mathbf{E} \cdot d\mathbf{A} = \iiint_R \rho dV,$$

where the **left-hand side** is the *boundary flux* and the **right-hand side** is the *interior content*.

The observer measures only the left-hand side; the right-hand side is inferred. This asymmetry — *boundary accessible, interior hidden* — is the key to the reinterpretation below.

1. State vector $|\psi\rangle$ — the outward-facing boundary pattern

In Gauss's law, the boundary flux

$$\iint_{\partial R} \mathbf{E} \cdot d\mathbf{A}$$

is the only quantity the observer can measure. The detailed interior charge distribution is inaccessible; only its outward effect is available.

The quantum state $|\psi\rangle$ can be understood in the same spirit.

It is not a literal map of the interior configuration of a system.

It is the *exterior footprint* of that configuration — the pattern of information that reaches the boundary and becomes available to the outside world. Many different microscopic arrangements can produce the same outward signature, just as many charge distributions yield identical flux. The state vector is therefore a *boundary summary*, not a microscopic description.

2. Bra $\langle\phi|$ — the observer's expected boundary pattern

In classical inference, one often compares a measured flux to a hypothetical flux pattern to test a hypothesis about the interior. The observer imagines what the boundary flux *would* look like if a certain interior configuration were present.

The bra plays exactly this role.

It represents the observer's *test pattern*, the specific boundary signature they are prepared to detect. It encodes a question posed to the boundary:

“Does the incoming boundary pattern match this one?”

The bra is therefore the **expected flux pattern**.

3. Bracket $\langle\phi|\psi\rangle$ — the match between expectation and receipt

The bracket is the quantum analogue of comparing:

- **left-hand side:** the observer's expected flux pattern,
- **right-hand side:** the actual boundary flux produced by the interior.

In Gauss's law, the left-hand side

$$\iint_{\partial R} \mathbf{E} \cdot d\mathbf{A}$$

is what the observer measures, and the right-hand side

$$\iiint_R \rho dV$$

is what the observer infers.

In the bracket, the **left-hand side** $\langle\phi|$ is the *expectation*, and the **right-hand side** $|\psi\rangle$ is the *receipt*. The bracket quantifies how well the two align.

This gives the bracket a clear operational meaning:
it is a boundary-level test of compatibility.

4. Born probability $|\langle\phi|\psi\rangle|^2$ — the strength of boundary agreement

In Gauss-style reasoning, the degree to which a measured flux matches an expected pattern determines how strongly the observer can support a hypothesis about the interior.

The Born rule expresses the same idea.

The probability is the squared magnitude of the match between expectation and receipt. It quantifies how strongly the boundary supports the observer’s hypothesis. Probability becomes a measure of *boundary agreement*, not an intrinsic randomness of nature.

5. Superposition — boundary ambiguity, not interior multiplicity

Gauss’s law teaches that many different interior charge distributions can produce the same outward flux. The boundary does not reveal which interior arrangement is present; it only reveals what is compatible with the observed flux.

Superposition can be understood the same way.

A superposed quantum state reflects the fact that the boundary signature is compatible with multiple interior possibilities. The system is not “in many states at once”; rather, the observer’s boundary information is insufficient to distinguish among several interior configurations.

Superposition becomes a statement about *what the boundary does not distinguish*.

6. Entanglement — shared boundary constraints

In classical settings, the flux through one part of a surface can constrain what must be happening elsewhere. The boundary behaves as a single, coupled ob-

ject. The outward flux is not a sum of independent contributions; it is a joint constraint.

Entanglement mirrors this structure.

Two subsystems share a boundary-compatible pattern that cannot be decomposed into independent contributions. Their outward signatures are linked by joint constraints on the interior. Entanglement becomes a statement about *shared boundary structure*, not nonlocal influence.

7. Collapse — committing to one boundary-compatible interpretation

Once an observer commits to a particular interpretation of the boundary flux, all incompatible interior possibilities are discarded. This commitment is irreversible because the mapping from interior to boundary is many-to-one: once interior distinctions collapse into the same boundary signature, they cannot be recovered.

Quantum collapse can be read the same way.

It is not a physical event inside the system, but the observer selecting one boundary-compatible interpretation and discarding the rest. Collapse becomes a shift in the observer’s boundary hypothesis.

8. Irreversibility — loss of interior detail at the boundary

Gauss’s law compresses interior information into a single outward flux. Distinct interiors can yield identical flux, so the inference cannot be reversed. This is a structural, not dynamical, irreversibility.

Quantum measurement inherits the same structure.

Once interior distinctions collapse into the same boundary signature, they cannot be reconstructed. The arrow of time arises from this information compression at the boundary.

9. Boundary-Based Understanding of Measurement, Collapse, and Quantum Paradoxes

The Gauss-style reinterpretation also sheds light on several long-standing quantum paradoxes without requiring any details of the UniNet model. Many of the classic puzzles—Copenhagen collapse, Wigner’s friend, Schrödinger’s cat, and the Einstein–Podolsky–Rosen argument—arise from treating the wavefunction as an interior object and collapse as an interior event. In the Gauss view, the wavefunction is a boundary signature, collapse is a boundary-level update of the observer’s hypothesis, and entanglement is a shared boundary constraint rather

than an interior connection. Once this interior–boundary distinction is made explicit, the paradoxes lose their force: no observer ever accesses the interior directly, so there is no contradiction in different observers maintaining different boundary-compatible descriptions.

9.1 Measurement as Boundary Inference

An observer approaches a system with a specific expectation of what they might detect at the boundary. When the actual boundary signature arrives, they compare it to that expectation. The bracket $\langle\phi|\psi\rangle$ expresses how well the observed pattern aligns with the one the observer was prepared to detect. Nothing inside the system is accessed; the entire measurement process is a comparison between an expected boundary pattern and the one that is actually received.

9.2 Collapse as Boundary Commitment

After observing a boundary signature, several interior configurations may still be compatible with it. The observer then selects one interpretation and discards the others. Because different interior possibilities can lead to the same outward-facing pattern, this choice cannot be undone. Collapse is this act of committing to a single boundary-compatible interpretation. The irreversibility of the interior-to-boundary mapping naturally gives this process a direction, even though the underlying quantum dynamics remain time-symmetric.

9.3 Entanglement as Joint Boundary Constraint

When two subsystems share a boundary structure, their outward-facing signatures cannot be treated independently. A constraint on one side restricts what must be true on the other. Entanglement reflects this shared boundary condition: the composite system presents a joint signature that cannot be decomposed into separate contributions. The correlations arise from the structure of the shared boundary, not from any influence passing between interiors.

9.4 Decoherence as Boundary Coarsening

A system’s interior may contain fine-grained distinctions, but the boundary does not always preserve them. As interactions accumulate, the boundary becomes less sensitive to subtle interior differences. Distinct interior configurations begin to produce indistinguishable outward signatures. Decoherence is this loss of boundary resolution: the interior may remain detailed, but the boundary no longer reflects those details, and the observer’s effective description becomes correspondingly simpler.

9.5 Wigner’s Friend as Nested Boundaries

Different observers interact with different boundaries, and therefore receive different outward-facing signatures. The friend inside the laboratory interacts with

the system’s immediate boundary and forms a description based on that access. Wigner, outside the laboratory, interacts with a larger boundary that includes both the friend and the system. Their descriptions differ without contradiction because they are based on different accessible regions.

This picture aligns naturally with the relativistic structure of lightcones. Each observer can only incorporate information that lies within their past lightcone, and the friend’s observations remain outside Wigner’s lightcone until the laboratory is opened. As long as their lightcones do not overlap, the two observers have access to different boundary data and therefore maintain different, boundary-compatible descriptions. Once causal contact becomes possible and their lightcones intersect, the boundary information available to both observers becomes shared, and their descriptions converge. The apparent paradox dissolves once it is recognized that no observer accesses the interior directly and that agreement is only required when their accessible-information regions—defined both by boundaries and by lightcones—finally connect.

9.6 Schrödinger’s Cat as Boundary Ambiguity

From outside the closed box, the observer sees only a boundary signature that is compatible with multiple interior possibilities. The cat is not simultaneously alive and dead; rather, the boundary does not distinguish between those possibilities. The superposition reflects the observer’s limited access, not a literal multiplicity inside the box. When the boundary is opened, the ambiguity resolves because the observer gains access to a more informative boundary.

9.7 Delayed-choice and quantum-eraser

Delayed-choice and quantum-eraser experiments often appear to suggest that future measurements influence past events. In the Gauss-style view, this impression disappears. The interior path of a particle is never accessible; only the boundary signature is. Changing the measurement apparatus changes the boundary conditions and therefore the outward-facing pattern that becomes available to the observer. The observer’s final inference reflects these boundary conditions, not a retroactive change to the interior history. The apparent “backwards-in-time” behavior arises from treating interior details as if they were observable, when in fact only the boundary ever is.

9.8 Delayed-Choice Experiments and Lightcone-Defined Boundaries

Delayed-choice and quantum-eraser experiments often appear to suggest that future measurements influence past events. In the Gauss-style view, this impression arises from treating interior details—such as which path a particle “took”—as if they were observable. The interior path is never directly accessible; only the boundary signature is. Changing the measurement apparatus changes the boundary conditions and therefore the outward-facing pattern that becomes available to the observer, not the interior history of the system.

This perspective aligns naturally with the relativistic structure of lightcones. Each observer can only incorporate information that lies within their past lightcone, and the configuration of the apparatus determines which boundary signatures can enter that lightcone. Before the final detection event, different observers may have access to different boundary data, and their descriptions need not agree. Once the relevant lightcones intersect—when the detection event occurs and the apparatus configuration becomes part of the observer’s accessible region—the boundary information becomes unified, and the observer commits to a single boundary-compatible interpretation.

The apparent retrocausal behavior in delayed-choice experiments therefore reflects a shift in boundary information, not a change to the interior past. The Gaussian view makes this explicit: the boundary determines what can be inferred, and the lightcone determines when that boundary information becomes available.

10. Gauss’s Law as the Continuum Limit of the UniNet Model

The interior–boundary logic developed in the previous sections was introduced entirely at the level of quantum notation. Nothing in that discussion relied on any particular physical substrate. However, the same structure first appeared in a much more concrete setting: the UniNet model, where the fundamental operation is to carve a graph into an interior and an exterior by performing a cut. In UniNet, every observable associated with a region must be expressible in terms of quantities defined on the boundary edges that separate the region from the rest of the graph.

UniNet: From Axioms to a Foundational Framework for Emergent Physics

This cut-based structure leads directly to a discrete analogue of Gauss’s law. A region is defined by a set of nodes, its boundary consists of the edges connecting those nodes to the outside, and the total outward flow across that boundary equals the total content inside the region. The observer sees only the boundary flows; the interior content is inferred. This is the same interior–boundary asymmetry that Gauss’s law expresses in continuous form.

The connection to the classical divergence theorem emerged only after the UniNet structure was already in place. When the discrete cut relation was compared to the continuous Gauss identity, the correspondence became clear: the discrete rule is the combinatorial version of the smooth boundary–interior relation. In this sense, Gauss’s law can be viewed as the continuum limit of the boundary logic that UniNet makes explicit.

The Gaussian reinterpretation of quantum notation arose later, when the same interior–boundary asymmetry was recognized in the operational roles of bras, kets, amplitudes, and collapse. The purpose of this section is not to describe

UniNet in detail, but simply to note that the discrete Gauss structure is not hypothetical. It is a concrete example of a substrate where the interior-boundary distinction is explicit, and where the classical Gauss relation appears naturally as the smooth limit of a simpler combinatorial rule. Readers interested in the broader context may consult the UniNet preprint, but the Gaussian interpretation stands on its own.

11. Concluding Remarks

The Gauss-style reinterpretation presented here is not intended as a literal description of physical reality, nor as a replacement for existing interpretations of quantum mechanics. It is simply a conceptual lens—one that highlights the universal asymmetry between what an observer can access at a boundary and what must be inferred about an interior. By reading kets, bras, amplitudes, and collapse through this boundary-first structure, several long-standing quantum puzzles become less mysterious. Superposition appears as boundary ambiguity, collapse as boundary commitment, and entanglement as a shared constraint on outward-facing signatures. The discrete Gauss law and the UniNet-inspired graph-cut perspective show that this interior–boundary logic is not tied to any particular physical substrate. It is a structural pattern that recurs whenever a system has inaccessible degrees of freedom and an accessible interface. Readers may adopt or discard this lens as they see fit; its purpose is simply to offer a compact and intuitive way of understanding the operational content of quantum notation.

At first glance, the structural parallels between Gauss-style interior–boundary mappings and several core elements of quantum theory—such as the Born rule, the bracket structure, and even unitary evolution—suggest that deeper mathematical correspondences may exist. Whether these similarities can be developed into formal equivalences is an open question. Establishing such results would require a more detailed functional-analytic treatment of boundary operators and quadratic evaluation maps, which lies beyond the scope of this note. The present work merely highlights the possibility and leaves the formal development to readers with a more mathematical inclination.

A more detailed discussion of the broader context in which this perspective arose, including its connection to the UniNet substrate model, can be found in the accompanying preprint archived on Zenodo.